

Unit 3 Notes

Combinatorics

Counting, recurrence relations,
and generating functions

These notes begin with elementary counting and build toward systematic methods for solving recurrence relations and extracting coefficients from generating functions. Each section explains the notation, develops the reasoning, and supplies practice with a worked solution.

The coverage follows the supplied syllabus: the pigeonhole principle; homogeneous and nonhomogeneous linear recurrence relations with constant coefficients; applications of recurrence relations; and generating functions. Counting prerequisites are introduced where they support these topics.

There are 52 numbered teaching sections, followed through to cumulative questions and detailed solutions. All diagrams are drawn in black and white. Work through the examples in sequence, and attempt the practice before reading its solution.

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1 Counting as a mathematical model

Combinatorics studies discrete arrangements, selections, and structures. Before applying a formula, identify what counts as one outcome, which outcomes are considered different, and what restrictions are imposed. Most counting mistakes begin with a mismatch between the formula and the objects being counted.

For example, selecting two students for an unordered committee is different from assigning a president and a secretary. In the committee, exchanging the students produces the same selection. In the officer assignment, exchanging them changes the outcome.

Main idea. A counting argument should explain a correspondence. If a procedure produces each permitted outcome exactly once, counting the choices in that procedure counts the objects. If outcomes appear repeatedly or some cannot be produced, the procedure must be corrected.

We use $|S|$ for the number of elements in a finite set S . The notation $\{1, 2, \dots, n\}$ denotes the integers from 1 through n . In sequence discussions, $\mathbb{N}$ will mean the nonnegative integers $0, 1, 2, \dots$; a different starting index will be stated explicitly.

Worked example. How many two-symbol strings can be made from $\{A, B, C\}$ when repetition is allowed? There are three choices for each position, giving nine strings: $AA, AB, AC, BA, BB, BC, CA, CB, CC$. The strings AB and BA are distinct because position matters.

Practice. How many unordered two-element subsets can be selected from $\{A, B, C\}$? Why is the answer not nine?

Solution. The subsets are $\{A, B\}$, $\{A, C\}$, and $\{B, C\}$, so there are three. A subset cannot repeat an element, and its order is irrelevant. These two changes remove both the repeated-letter strings and the distinction between opposite orders.

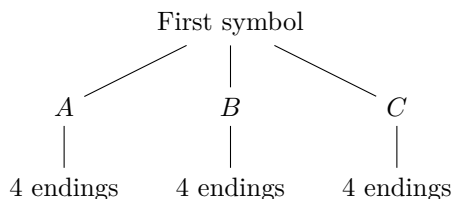
Common error. State whether repetition is permitted and whether order matters before calculating. These are properties of the problem, not optional settings chosen after a formula has been selected.

2 The sum and product principles

The sum principle counts disjoint alternatives. If one class has a outcomes and another disjoint class has b , choosing an outcome from either class gives $a + b$ possibilities. Disjointness matters because an overlapping outcome would otherwise be counted twice.

The product principle counts stages. If there are a choices for the first stage and exactly b choices for the second after each first choice, there are ab complete outcomes. The available second-stage objects may change with the first choice; what matters for this simple product is that their number stays b .

Worked example. A code uses a letter from $\{A, B, C\}$ followed by a digit from $\{1, 2, 3, 4\}$. There are $3 \cdot 4 = 12$ codes. If a different allowed code consists only of one digit, these four additional codes are disjoint from the two-symbol codes, giving $12 + 4 = 16$ in total.



Practice. A student chooses one of three mathematics courses and one of two computing courses. A particular mathematics course conflicts with one particular computing course. How many valid pairs remain?

Solution. Without the restriction there are $3 \cdot 2 = 6$ pairs. Exactly one pair is forbidden, so five remain. We cannot multiply by a uniform number of allowed computing choices after every mathematics choice: one branch has one allowed choice while the other two have two. Adding branch counts, $1 + 2 + 2 = 5$, gives the same result.

Common error. “Or” does not automatically mean add, and “and” does not automatically mean multiply. Check overlap for alternatives and check the dependence of choices for stages.

3 Factorials and binomial coefficients

The factorial $n!$ means $n(n-1)\cdots 2\cdot 1$, with $0! = 1$. It counts arrangements of n distinct objects: choose the first in n ways, the second in $n-1$, and continue until only one remains.

The binomial coefficient $\binom{n}{r}$ counts unordered selections of r distinct objects from n , where $0 \leq r \leq n$. Its formula is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

The numerator divided by $(n-r)!$ counts ordered selections. Each unordered selection appears in $r!$ orders, so dividing by $r!$ removes precisely that duplication.

Worked example. Choosing three representatives from eight people gives $\binom{8}{3} = 56$. Assigning three distinct positions among the eight gives $8 \cdot 7 \cdot 6 = 336$. The factor of $3! = 6$ between the answers records the six orders of each selected triple.

These coefficients will reappear in generating functions. The coefficient of x^r in $(1+x)^n$ is $\binom{n}{r}$: from each of n factors, choose either 1 or x , and choose the x term in exactly r factors.

Practice. How many length-five binary strings contain exactly two 1s?

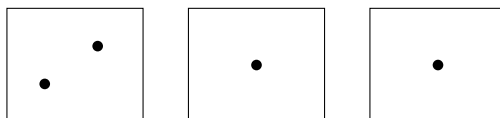
Solution. Choose the two positions that contain 1; all remaining positions must contain 0. There are $\binom{5}{2} = 10$ choices. No extra factor is needed for the zeros or the order of the ones, because positions already determine the entire string.

Common error. The formula $\binom{n}{r}$ assumes selection without repetition from distinct objects. It does not directly count unrestricted selections of repeated coin denominations. Generating functions will handle those situations by representing the permitted multiplicities explicitly.

4 The pigeonhole principle

If more than k objects are placed into k boxes, at least one box contains two or more objects. The objects are traditionally called pigeons and the categories holes, but in applications the categories may be remainders, birthdays, intervals, or possible data values.

Main idea. If every box held at most one object, all boxes together could hold at most k . Having more than k objects makes that assumption impossible. The proof is a capacity argument.



Four objects placed in three boxes

Worked example. Among 13 people, at least two were born in the same month. The people are the objects and the 12 calendar months are the boxes. We need not know their actual birthdays or which month repeats.

Practice. Prove that among any 11 integers, two have the same last decimal digit.

Solution. There are ten possible final digits, 0 through 9. Classify each integer by its residue modulo 10; this avoids ambiguity with negative-number notation. Eleven integers placed into ten residue classes force two into the same class. Their difference is therefore divisible by 10.

Common error. The principle guarantees existence, not the identity of the repeated category. It also does not require equal-sized or equally likely categories. Probability plays no role in the argument.

The main creative step is choosing useful boxes. Once every object has been assigned to exactly one of a known number of categories, the conclusion usually follows immediately.

5 The generalised pigeonhole principle

If N objects are placed into k nonempty-indexed boxes, with $k \geq 1$, some box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects. The ceiling $\lceil t \rceil$ is the smallest integer greater than or equal to the real number t . For example, $\lceil 17/5 \rceil = 4$.

To understand the formula, compare the most crowded box with the average occupancy N/k . If every box were below that average, the total would be less than N . Since occupancies are integers, at least one must reach the ceiling of the average.

Worked example. With 25 students in six tutorial groups, some group has at least $\lceil 25/6 \rceil = 5$ students. This does not say that every group has five students or that some group has exactly five. One group might contain many more.

There is a dual observation: some box contains at most $\lfloor N/k \rfloor$ objects, where the floor is the greatest integer at most its input. This follows by comparing the least crowded box with the average.

Practice. In a set of 73 people, show that at least seven share a birth month. Could the same argument guarantee eight?

Solution. The lower bound is $\lceil 73/12 \rceil = 7$, so seven are guaranteed. Eight are not forced: distributing the people as one group of seven and eleven groups of six uses $7 + 66 = 73$ people with no month containing eight. This distribution also shows that the bound is sharp for the available information.

Common error. Round upward when bounding the maximum occupancy from below. Rounding $73/12$ down would give a true but weaker guarantee and miss the strongest conclusion supplied by the principle.

6 Threshold questions and worst-case distributions

To guarantee at least r objects in one of k boxes, the smallest sufficient total is

$$k(r - 1) + 1.$$

The reason is concrete: a distribution avoiding r in every box may place at most $r - 1$ in each, for a total of at most $k(r - 1)$. One additional object makes avoidance impossible.

This method answers “how many are needed to guarantee” questions. It begins by constructing the largest possible unsuccessful arrangement rather than guessing a successful one.

Worked example. How many people guarantee that five have the same birth month? The largest arrangement with at most four per month has $12 \cdot 4 = 48$ people. The next person forces a fifth in some month, so the answer is 49.

Practice. Socks come in four colours, with at least ten of each available. How many blind selections guarantee three socks of one colour? How many guarantee three socks of a specified colour, if there is no fixed bound on the quantities of the other colours?

Solution. For any colour, at most two of each colour can be drawn without a triple. That gives eight unsuccessful draws, so nine guarantee a triple. For a specified colour, no finite bound follows from the given unbounded supply of the other colours: a selection could keep drawing those colours. The second question therefore needs additional inventory information.

Common error. A guarantee of “some category” is not a guarantee about a prescribed category. Also respect finite supplies: if a box cannot contain $r - 1$ objects because fewer are available, the worst-case calculation must use that smaller capacity.

When capacities vary, the same reasoning becomes $c_1 + \cdots + c_k + 1$, where c_i is the maximum permitted occupancy of box i while still avoiding the desired event.

7 Remainders as pigeonholes

For a positive integer m , every integer has exactly one remainder among $0, 1, \dots, m-1$ upon division by m . These m residue classes provide a natural finite set of boxes, even when the integers themselves are arbitrarily large.

Two integers with the same remainder differ by a multiple of m . Indeed, if $a = mu + r$ and $b = mv + r$, then $a - b = m(u - v)$. We write $a \equiv b \pmod{m}$ for this relationship.

Worked example. Among any $m + 1$ integers, two have a difference divisible by m . Put each integer into its remainder class modulo m . The pigeonhole principle gives a repeated class, and the displayed calculation converts the shared category into the desired divisibility statement.

Practice. Prove that from any six integers, two can be selected whose difference is divisible by 5. Does the same argument prove that two have a sum divisible by 5?

Solution. The difference claim follows from six objects and five remainder classes. The same-class argument does not establish the sum claim: two numbers with remainder 1 have sum with remainder 2. In fact, the sum claim is false without further assumptions; six integers all congruent to 1 modulo 5 have no pair whose sum is divisible by 5.

Common error. Select categories according to the desired algebraic conclusion. Equal remainders control differences. Complementary remainders may help control sums, but they require a different grouping and often different hypotheses.

The use of residues is a compression of information. Instead of tracking the entire integer, retain only the part that matters for divisibility by the chosen modulus. A good pigeonhole proof discards irrelevant detail while preserving the feature needed in the conclusion.

8 Partial sums and a divisible consecutive block

Pigeonhole arguments can act on derived objects rather than the original data. Partial sums are especially useful when a conclusion concerns a consecutive block of terms, because a block sum is the difference of two partial sums.

Worked example. Given any m integers $a_1, \dots, a_m$, there is a nonempty consecutive block whose sum is divisible by m . Define

$$S_0 = 0, \quad S_j = a_1 + \dots + a_j \quad (1 \leq j \leq m).$$

There are $m + 1$ partial sums and only m remainder classes modulo m . Therefore S_i and S_j have the same remainder for some $i < j$. Their difference is

$$S_j - S_i = a_{i+1} + \dots + a_j,$$

which is divisible by m . Since $i < j$, the block is nonempty.

Practice. Find such a block in the list 2, 5, 1, 4 with $m = 4$, using partial sums.

Solution. The partial sums are 0, 2, 7, 8, 12, whose remainders modulo 4 are 0, 2, 3, 0, 0. The equal remainders of S_0 and S_3 give the block $2 + 5 + 1 = 8$. The equal remainders of S_3 and S_4 give the one-term block 4. Either block is a valid answer; the existence theorem does not claim uniqueness.

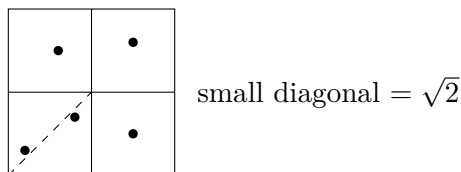
Common error. Include $S_0 = 0$. It allows a block starting with the first term to appear as a difference just like every other block, and it gives exactly the extra object required by the pigeonhole principle.

The proof works with negative integers as well as positive ones. Positivity is not needed because the argument concerns remainders and differences, not monotonicity of the partial sums.

9 Geometric pigeonhole arguments

Geometry supplies boxes by partitioning a region. If two points must fall in the same small region, a bound on that region's diameter gives a bound on their distance. The *diameter* is the largest possible distance between two points of the region.

Worked example. Place five points in a square of side length 2. Divide the square into four unit squares. At least two points lie in the same small square, and their distance is at most its diagonal, $\sqrt{2}$.



Boundary points need a consistent assignment rule, such as assigning an internal dividing-line point to the square on its right or above. The essential requirement is that every point belongs to exactly one box for counting purposes. Every assigned square still has diagonal $\sqrt{2}$.

Practice. Show that among ten points in a unit square, two have distance at most $\sqrt{2}/3$.

Solution. Divide the square into a 3 by 3 grid. There are nine squares of side $1/3$, so ten points force two into one small square. Its diagonal is $\sqrt{(1/3)^2 + (1/3)^2} = \sqrt{2}/3$, giving the required bound.

Common error. The diagonal bounds Euclidean distance; the side length generally does not. Two points near opposite corners of the same small square can be farther apart than its side length.

These arguments provide guaranteed bounds. A partition may not give the best possible geometric constant, so do not describe a bound as optimal unless sharpness has also been established.

10 Designing boxes from a hidden structure

Sometimes the right categories are not obvious from the statement. A useful strategy is to remove one feature of each object until a simpler representative remains. If equal representatives imply the desired relation, they can serve as boxes.

Worked example. Select $n + 1$ distinct integers from $\{1, 2, \dots, 2n\}$. Then one selected integer divides another. Every positive integer can be written uniquely as $2^k u$, where u is odd and $k \geq 0$: repeatedly divide by 2 until an odd number remains. Use this odd part u as the box label.

There are only n possible odd labels among $1, 3, \dots, 2n - 1$. Two selected numbers therefore have the same odd part, say $2^a u$ and $2^b u$. Since the numbers are distinct, $a \neq b$. If $a < b$, then $2^b u = 2^{b-a}(2^a u)$, so the smaller power-of-two multiple divides the larger.

Practice. Among $2, 3, 4, 5, 6, 7$ selected from $\{1, \dots, 10\}$, group the numbers by odd part and identify a dividing pair.

Solution. The odd parts are $1, 3, 1, 5, 3, 7$. Thus 2 and 4 share odd part 1, and 3 and 6 share odd part 3. We obtain the pairs $2 \mid 4$ and $3 \mid 6$. The theorem promises at least one, and this selection happens to supply two.

Common error. Distinctness is part of this theorem's useful conclusion. If repeated selections were allowed, identical numbers would trivially divide each other, changing the nature of the problem. Always keep the original object definition visible while constructing categories.

The proof illustrates a general tactic: identify an invariant, meaning a feature that stays unchanged under a relevant transformation. Here multiplying by 2 leaves the odd part unchanged, and shared odd parts reveal a divisibility chain.

11 Sharp bounds and the limitations of counting guarantees

A lower bound is *sharp* when an example achieves it, showing that it cannot be improved under the given assumptions. A guarantee proof and a sharpness example serve different purposes: the proof covers every arrangement, while the example blocks a stronger universal claim.

Worked example. With 17 objects in five boxes, at least four occupy some box. The distribution $(4, 4, 3, 3, 3)$ has total 17 and no occupancy above four. Therefore a guarantee of five would be false, and the bound four is sharp.

In the divisibility theorem from the previous section, $n + 1$ chosen integers suffice. The threshold cannot be reduced to n : the set $\{n + 1, n + 2, \dots, 2n\}$ has n elements and no distinct pair with one dividing the other. Any larger multiple of one of these elements exceeds $2n$.

Practice. Find the smallest number of people that guarantees six share a birth month, and justify that the number is minimal.

Solution. At most five per month gives $12 \cdot 5 = 60$ people without six in a month. Thus 61 suffice. The distribution of exactly five people in each month shows that 60 do not suffice. Both parts are needed to justify the word “smallest.”

Common error. An argument that 61 works establishes sufficiency, not minimality. A minimal threshold requires an unsuccessful arrangement immediately below the proposed threshold, or another lower-bound argument excluding every smaller candidate.

Pigeonhole reasoning says little about where the crowded box is or how other boxes are populated. It remains powerful precisely because it needs so little information: a total count and a limited number of categories can force a structural conclusion.

12 Pigeonhole practice with complete solutions

Practice.

1. Among 31 integers, prove that four have the same remainder modulo 10.
2. How many draws from five categories guarantee four objects in some category?
3. Show that among seven real numbers in $[0, 1]$, two differ by at most $1/6$.

Solution. 1. There are ten possible remainders. The generalised principle gives a box with at least $\lceil 31/10 \rceil = 4$ members. The conclusion is about a shared remainder, and hence all pairwise differences within that group are divisible by 10.

2. The largest distribution avoiding four has three objects in each of five categories, for 15 objects. Therefore 16 guarantee four. The arrangement $(3, 3, 3, 3, 3)$ shows minimality if supplies allow it.

3. Divide $[0, 1]$ into six intervals of length $1/6$, using half-open intervals except for the final one, which includes 1. Every point is assigned to exactly one interval. Seven numbers force two into one interval, and the difference of two numbers in such an interval has absolute value at most $1/6$.

A second solution to 3. Sort the numbers as $x_1 \leq \dots \leq x_7$. The six adjacent gaps sum to $x_7 - x_1 \leq 1$. If all gaps exceeded $1/6$, their sum would exceed 1, a contradiction. Hence some adjacent pair has the required distance. This is the same capacity idea expressed through gaps rather than interval boxes.

Common error. In a proof about real numbers, distinguish the signed difference $a - b$ from the distance $|a - b|$. A statement that two numbers “differ by at most” a bound normally concerns the absolute value.

Before moving to recurrences, explain the common structure of all three solutions: identify finitely many places in which the relevant objects or total size can be distributed, then show that uniform avoidance would violate the total.

13 Sequences, recurrence relations, and initial values

A sequence is an indexed list $a_0, a_1, a_2, \dots$. The term a_n is the value at index n ; the subscript is a label, not multiplication. A *recurrence relation* expresses a term using earlier terms and possibly the index.

For example, $a_n = 2a_{n-1} + 1$ for $n \geq 1$ gives an update rule. An initial value such as $a_0 = 0$ selects the particular sequence $0, 1, 3, 7, 15, \dots$. Without a_0 , many sequences satisfy the same rule.

A recurrence has *order* k if the farthest previous term involved is a_{n-k} with a nonzero coefficient. It is *linear* when sequence terms occur only to the first power and are not multiplied together. It has *constant coefficients* when those coefficients do not depend on n .

Worked example. $a_n = 3a_{n-1} - 2a_{n-2} + n$ is second order, linear, nonhomogeneous, and has constant coefficients. The additional term n is the forcing term. By contrast, $a_n = (a_{n-1})^2 + 1$ is nonlinear, and $a_n = na_{n-1}$ has a variable coefficient.

Practice. Classify $b_n = 4b_{n-1}$ and $c_n = c_{n-1} + c_{n-2} + 2^n$. How many initial values are normally needed?

Solution. The first is first-order linear homogeneous with constant coefficient 4, and needs one initial value. The second is second-order linear nonhomogeneous with constant coefficients and needs two initial values. “Homogeneous” means no independent forcing term remains; the presence of 2^n makes the second nonhomogeneous.

Common error. Always record where the recurrence starts. A second-order rule is usually imposed for $n \geq 2$ when a_0, a_1 are given. Applying it at an earlier index may introduce undefined terms.

14 Building a recurrence from a description

Before solving a recurrence, justify that it models the intended process. Define the quantity being counted, divide the possible objects by a meaningful final step, and show that the cases are disjoint and exhaustive.

Worked example. Let t_n count ways to tile a $1 \times n$ strip using 1×1 squares and 1×2 dominoes. A tiling ends either in a square, leaving a tiling of length $n - 1$, or in a domino, leaving one of length $n - 2$. These cases are disjoint and cover every tiling. Therefore

$$t_n = t_{n-1} + t_{n-2} \quad (n \geq 2).$$

The initial values are $t_0 = 1$ and $t_1 = 1$. There is one empty tiling of an empty strip; this convention lets a domino by itself be counted through the t_0 contribution when $n = 2$.

length $n - 1$	1
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length $n - 2$	2
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Practice. Find t_2, t_3, t_4 and explain why the recurrence does not multiply the two preceding counts.

Solution. The values are 2, 3, 5. We add because a completed tiling belongs to one of two alternative final-piece cases. We are not independently choosing both a length- $(n - 1)$ tiling and a length- $(n - 2)$ tiling to make one object, so the product principle does not apply.

Common error. A recurrence guessed from several initial values is not yet a justified model. Explain the structural correspondence that makes it hold for every permitted n .

15 First-order homogeneous recurrences

Consider $a_n = ra_{n-1}$ for $n \geq 1$, with $a_0 = A$. Each step multiplies the previous value by the same constant r . Repeated substitution gives

$$a_1 = rA, \quad a_2 = r^2A, \quad a_3 = r^3A,$$

and in general $a_n = Ar^n$ when $r \neq 0$. If $r = 0$, then $a_0 = A$ and every later term is zero; this can be stated separately without needing a convention for 0^0 .

Main idea. A recurrence is a local rule, while a closed formula gives a term directly from its index. A correct closed formula must satisfy both the recurrence and the initial value.

Worked example. Solve $a_n = -2a_{n-1}$ with $a_0 = 3$. The solution is $a_n = 3(-2)^n$. The negative multiplier causes alternating signs: $3, -6, 12, -24, \dots$. Its magnitude doubles at each step.

Practice. A sequence satisfies $b_n = \frac{1}{2}b_{n-1}$ with $b_0 = 80$. Find b_n and the first index for which $b_n < 6$.

Solution. The formula is $b_n = 80/2^n$. The early values are $80, 40, 20, 10, 5$. Thus the first such index is $n = 4$. The answer concerns strict inequality; if the threshold were exactly 5, the first term strictly below it would occur at $n = 5$.

Common error. If the initial value is $a_1 = A$ instead of $a_0 = A$, the formula becomes $a_n = Ar^{n-1}$. An index shift changes the exponent even though the underlying multiplier is unchanged.

This simple family motivates the characteristic-root method. Powers r^n are natural candidates because shifting the index by one changes them only by a constant factor, matching the structure of a constant-coefficient recurrence.

16 First-order forcing and repeated substitution

For $a_n = ra_{n-1} + g_n$, the independent contribution g_n enters at step n and is multiplied by r at each later step. Repeated substitution gives

$$a_n = r^n a_0 + \sum_{j=1}^n r^{n-j} g_j.$$

The index j identifies when a contribution entered. The exponent $n - j$ counts how many subsequent multiplications it experiences. This formula is useful even when g_n is not a simple polynomial or exponential.

Worked example. If $a_n = 2a_{n-1} + 3$ and $a_0 = 1$, then

$$\begin{aligned} a_n &= 2^n + 3(1 + 2 + \cdots + 2^{n-1}) \\ &= 2^n + 3(2^n - 1) \\ &= 4 \cdot 2^n - 3. \end{aligned}$$

Checking $n = 0$ gives 1. Substituting the closed formula into the recurrence verifies every later value.

For a constant forcing $g_n = c$ and $r \neq 1$, the finite geometric sum gives

$$a_n = r^n a_0 + c \frac{r^n - 1}{r - 1}.$$

If $r = 1$, the recurrence simply adds c repeatedly, so $a_n = a_0 + nc$. The separate case avoids dividing by zero.

Practice. Solve $b_n = b_{n-1} + 2n$ for $n \geq 1$, with $b_0 = 5$.

Solution. Adding the increments gives $b_n = 5 + 2 \sum_{j=1}^n j = 5 + n(n+1)$. The first values are 5, 7, 11, 17, which agree with the formula. The forcing changes with the step number, so it cannot be replaced by a fixed increment.

Common error. In the summation formula, the newest input g_n is multiplied by $r^0 = 1$, not by r . Drawing two or three update steps before writing the general sum prevents this common timing error.

17 Why a characteristic equation appears

A second-order homogeneous recurrence has the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} \quad (n \geq 2).$$

Assume for now that $c_2 \neq 0$, so the recurrence genuinely has order two. Try a solution $a_n = r^n$. Substitution gives $r^n = c_1 r^{n-1} + c_2 r^{n-2}$. For a nonzero candidate r , cancelling r^{n-2} yields

$$r^2 - c_1 r - c_2 = 0.$$

This is the *characteristic equation*. A root is a multiplier whose power sequence is preserved by the recurrence.

Main idea. Because the recurrence is linear and homogeneous, a sum of solutions is a solution, and a constant multiple of a solution is a solution. Thus characteristic roots supply building blocks that can be combined to meet initial values.

Worked example. For $a_n = 5a_{n-1} - 6a_{n-2}$, the equation is $r^2 - 5r + 6 = 0$, with roots 2, 3. Both 2^n and 3^n satisfy the update rule. Therefore $A2^n + B3^n$ also satisfies it for any constants A, B .

Practice. Write the characteristic equation for $b_n = 4b_{n-1} + 5b_{n-2}$ and find its roots.

Solution. Moving all terms to the left gives $r^2 - 4r - 5 = 0 = (r - 5)(r + 1)$. The roots are 5 and -1 . The sign of the constant term is negative because the recurrence has $+5b_{n-2}$ on the right.

Common error. Form the polynomial after moving the recurrence into a consistent left-side form. Sign errors at this stage produce a different recurrence and invalidate every later calculation.

If the last coefficient is zero, reduce the true order first or treat the degenerate case separately. This avoids relying on cancellation involving a zero characteristic root.

18 Distinct characteristic roots

If a second-order homogeneous recurrence has two distinct characteristic roots r_1, r_2 , its general solution is

$$a_n = Ar_1^n + Br_2^n.$$

The constants A, B are fixed by the two initial conditions. Distinct roots make the two power sequences independent, so they can represent all solutions of the second-order recurrence.

Worked example. Solve $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 2, a_1 = 5$. The roots are 2, 3, so $a_n = A2^n + B3^n$. The initial conditions give

$$A + B = 2, \quad 2A + 3B = 5.$$

Subtract twice the first equation from the second to get $B = 1$, then $A = 1$. Hence $a_n = 2^n + 3^n$. The recurrence predicts $a_2 = 25 - 12 = 13$, agreeing with $4 + 9 = 13$.

Practice. Solve $b_n = 3b_{n-1} - 2b_{n-2}$ with $b_0 = 0, b_1 = 1$.

Solution. The characteristic polynomial is $(r - 1)(r - 2)$, so $b_n = A + B2^n$. At $n = 0$, $A + B = 0$; at $n = 1$, $A + 2B = 1$. Therefore $B = 1, A = -1$, giving $b_n = 2^n - 1$. The next value is 3, consistent with the recurrence.

Common error. Solve for the coefficients after finding the general form. Guessing a coefficient from only one initial value can accidentally ignore the other independent component. Also keep a term whose coefficient later turns out to be zero until the initial conditions have actually established that fact.

There is a simple completeness reason: any two solutions with the same first two terms have the same third term by the recurrence, then the same fourth, and so on. Matching the two starting terms therefore identifies the whole sequence.

19 Repeated characteristic roots

When the characteristic polynomial has a repeated root r , the two expressions Ar^n and Br^n collapse into one, since their sum is $(A+B)r^n$. They cannot independently fit two initial values. The missing independent solution is nr^n .

Thus a double root produces

$$a_n = (A + Bn)r^n.$$

More generally, a root of multiplicity m contributes $r^n, nr^n, \dots, n^{m-1}r^n$. The polynomial factor records the extra independent solutions associated with repetition.

Worked example. Solve $a_n = 4a_{n-1} - 4a_{n-2}$ with $a_0 = 1, a_1 = 6$. The polynomial $(r-2)^2$ has a double root at 2, so $a_n = (A + Bn)2^n$. The first condition gives $A = 1$. The second gives $2(1 + B) = 6$, hence $B = 2$. Therefore $a_n = (1 + 2n)2^n$.

Practice. Solve $b_n = 2b_{n-1} - b_{n-2}$ with $b_0 = 3, b_1 = 8$.

Solution. The polynomial is $(r-1)^2$. The general solution is $A + Bn$. At $n = 0$, $A = 3$; at $n = 1$, $3 + B = 8$, so $B = 5$. Thus $b_n = 3 + 5n$. The recurrence says that consecutive differences are equal: $b_n - b_{n-1} = b_{n-1} - b_{n-2}$. This explains directly why arithmetic sequences appear.

Common error. A repeated root does not mean writing the same exponential twice. Multiplication by n is essential. For a triple root, the required factor is a general quadratic $A + Bn + Cn^2$, not merely $A + Bn$.

The repeated-root form also anticipates generating functions: repeated factors in a denominator produce polynomial factors in the coefficients. Later sections connect these two descriptions.

20 Complex roots and real sequences

A real-coefficient recurrence may have nonreal characteristic roots. They occur in conjugate pairs. If the pair is

$$r_{1,2} = \rho(\cos \theta \pm i \sin \theta),$$

then the corresponding real solutions can be written

$$a_n = \rho^n (C \cos(n\theta) + D \sin(n\theta)).$$

Here $i^2 = -1$, ρ is the root's magnitude, and θ is its angle. This form describes oscillation, possibly scaled by growth or decay.

Worked example. For $a_n = -a_{n-2}$, the characteristic equation is $r^2 + 1 = 0$, with roots $i, -i$. Thus $\rho = 1$ and $\theta = \pi/2$. With $a_0 = 1, a_1 = 0$, the solution is $a_n = \cos(n\pi/2)$, giving $1, 0, -1, 0, 1, 0, \dots$

You need not use complex numbers to generate the terms or verify this example. The recurrence itself shows the alternating two-step pattern. The complex form provides a systematic solution method for less obvious cases.

Practice. Solve the same recurrence with $b_0 = 0, b_1 = 2$.

Solution. In the real form $C \cos(n\pi/2) + D \sin(n\pi/2)$, the first condition gives $C = 0$, and the second gives $D = 2$. Hence $b_n = 2 \sin(n\pi/2)$, with values $0, 2, 0, -2, 0, 2, \dots$. Substituting shows that increasing n by two reverses the sign, as required.

Common error. Nonreal characteristic roots do not force nonreal sequence values. Conjugate components combine into a real expression when the recurrence and initial conditions are real. Do not discard complex roots simply because the original quantities are real numbers.

If trigonometric notation is unfamiliar, use this section as an extension after mastering distinct and repeated real roots; the remaining standard methods do not require complex arithmetic in their worked examples.

21 Higher-order homogeneous recurrences

An order- k linear homogeneous recurrence with constant coefficients has the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k},$$

with $c_k \neq 0$. Its characteristic polynomial is

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_k.$$

For every distinct root, include the exponential solution and enough powers of n to account for its multiplicity. The total number of independent coefficients is k , matching the number of initial conditions.

Worked example. For $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$, the polynomial factors as $(r-1)(r-2)(r-3)$. Hence $a_n = A + B2^n + C3^n$. If $a_0 = 3, a_1 = 6, a_2 = 14$, the choice $A = B = C = 1$ satisfies all three equations, giving $a_n = 1 + 2^n + 3^n$.

Practice. A characteristic polynomial is $(r-2)^2(r+1)$. State the general solution and write the corresponding recurrence.

Solution. The solution is $(A + Bn)2^n + C(-1)^n$. Expanding the polynomial gives

$$(r^2 - 4r + 4)(r + 1) = r^3 - 3r^2 + 4.$$

Therefore the recurrence is $a_n = 3a_{n-1} - 4a_{n-2} + 4a_{n-3}$ for $n \geq 3$. The coefficient of a_{n-2} is zero, but the recurrence still has order three because the a_{n-3} term is present.

Common error. Order is determined by the largest lag with nonzero coefficient, not the number of terms printed on the right. A sparse third-order recurrence can still require three initial values.

In practice, factoring the characteristic polynomial may be the difficult algebraic step. Always verify a proposed factorisation by expanding it, especially when signs alternate.

22 Initial conditions and uniqueness

An explicit order- k recurrence with specified forcing and k consecutive initial values determines at most one forward sequence. Once the starting values are known, the recurrence fixes the next value, and repeated application fixes every later value.

This observation provides a powerful verification method. If a proposed closed formula satisfies the recurrence for all required indices and agrees with the initial values, it must be the desired solution. Matching a few later terms alone is not enough, but symbolic recurrence verification plus initial agreement is enough.

Worked example. The expression $a_n = 2^n - 1$ agrees with $a_0 = 0, a_1 = 1$ and satisfies $a_n = 3a_{n-1} - 2a_{n-2}$ because

$$3(2^{n-1} - 1) - 2(2^{n-2} - 1) = 2^n - 1.$$

Uniqueness therefore establishes the formula for the entire sequence.

Practice. For $b_n = 2b_{n-1} + 1$ with $b_0 = 4$, someone proposes $b_n = 5 \cdot 2^n - 1$. Verify it completely.

Solution. At $n = 0$, the formula gives $5 - 1 = 4$. For every $n \geq 1$,

$$\begin{aligned} 2b_{n-1} + 1 &= 2(5 \cdot 2^{n-1} - 1) + 1 \\ &= 5 \cdot 2^n - 1 \\ &= b_n. \end{aligned}$$

The recurrence and initial value both hold, so the formula is correct by forward uniqueness.

Common error. A formula may satisfy the recurrence but have the wrong starting constants. Conversely, fitting the initial values does not establish the update rule. Both checks are essential.

Initial conditions given at nonconsecutive indices require separate care: they may fail to determine a unique solution or may be inconsistent. The straightforward uniqueness statement here assumes a complete consecutive starting block and a rule explicitly solving for the next term.

23 Nonhomogeneous recurrences: the general principle

Write a linear recurrence as $L[a]_n = g_n$, where L means the left-side combination of sequence terms. For example,

$$L[a]_n = a_n - 3a_{n-1} + 2a_{n-2}.$$

A *homogeneous solution* h_n satisfies $L[h]_n = 0$. A *particular solution* p_n satisfies $L[p]_n = g_n$. By linearity, $h_n + p_n$ satisfies the original recurrence.

Main idea. Every solution has the form

$$a_n = h_n + p_n.$$

Indeed, subtracting one particular solution from any solution leaves something whose forcing cancels to zero. This is why the characteristic-root method remains useful even when the recurrence is nonhomogeneous.

Worked example. For $a_n = 2a_{n-1} + 3$, the homogeneous solutions are $C2^n$. A constant particular solution $p_n = K$ must satisfy $K = 2K + 3$, giving $K = -3$. Therefore $a_n = C2^n - 3$. Only after combining these parts should the initial value be used to determine C .

Practice. Solve $b_n = 3b_{n-1} + 4$ with $b_0 = 2$ using this decomposition.

Solution. The homogeneous part is $C3^n$. A constant trial gives $K = 3K + 4$, hence $K = -2$. The complete form is $b_n = C3^n - 2$. Using $b_0 = 2$ gives $C = 4$, so $b_n = 4 \cdot 3^n - 2$.

Common error. A particular solution is not required to satisfy the initial values. Its role is to reproduce the forcing. The homogeneous part supplies the flexibility needed to fit the starting data. Trying to force initial values onto the particular part can make a solvable problem appear inconsistent.

Different particular solutions may look different. Their difference must be homogeneous, so after the arbitrary homogeneous constants are adjusted, they describe the same full family.

24 Constant forcing

When the forcing is constant, a constant particular solution is the first trial. It works unless constants already belong to the homogeneous solution space, which occurs when 1 is a characteristic root.

Worked example. Solve $a_n = 3a_{n-1} - 2a_{n-2} + 4$, with $a_0 = 0, a_1 = 1$. The homogeneous polynomial is $(r-1)(r-2)$, so a constant trial would be absorbed into the homogeneous part and yield zero forcing. Try $p_n = Kn$ instead. Then

$$Kn - 3K(n-1) + 2K(n-2) = -K.$$

Setting this equal to 4 gives $K = -4$. Thus $a_n = A + B2^n - 4n$. The initial equations are $A + B = 0$ and $A + 2B - 4 = 1$, yielding $B = 5, A = -5$. Hence $a_n = 5 \cdot 2^n - 4n - 5$.

Practice. Find a constant particular solution for $b_n = 2b_{n-1} + b_{n-2} + 6$.

Solution. With $p_n = K$, the recurrence requires $K = 2K + K + 6$. Thus $-2K = 6$ and $K = -3$. Constants are not homogeneous here, since the characteristic polynomial $r^2 - 2r - 1$ is nonzero at $r = 1$.

Common error. Failure of a constant trial does not mean the recurrence has no solution. It may mean that the trial overlaps the homogeneous family. Multiplying by n removes a simple overlap; a repeated root at 1 requires a higher power of n .

The forcing term describes what is injected at each update, while a particular solution describes a whole sequence whose differences reproduce those injections. It is therefore quite possible for positive forcing to have a negative particular solution, as the examples show.

25 Polynomial forcing

If the forcing is a polynomial in n of degree d , begin with a general polynomial particular solution of degree d . Include every lower-degree term because index shifts mix polynomial degrees. If 1 is a characteristic root of multiplicity s , multiply the whole trial polynomial by n^s .

Worked example. Solve $a_n = 2a_{n-1} + n$ with $a_0 = 0$. The homogeneous solution is $C2^n$, and 1 is not a characteristic root. Try $p_n = An + B$. Substitution into $p_n - 2p_{n-1} = n$ gives

$$An + B - 2(A(n-1) + B) = -An + (2A - B).$$

Matching coefficients with n yields $-A = 1$ and $2A - B = 0$, so $A = -1, B = -2$. Thus $a_n = C2^n - n - 2$. The initial value gives $C = 2$, resulting in $a_n = 2^{n+1} - n - 2$.

Practice. Find a suitable trial form, without solving its coefficients, for forcing $3n^2 - 5n + 7$ when (a) 1 is not a root; (b) 1 is a double root.

Solution. In (a), use $An^2 + Bn + C$. In (b), use $n^2(An^2 + Bn + C)$. The full quadratic is needed even though a leading-term guess might look simpler. The extra n^2 in (b) compensates for the double overlap at the exponential factor 1^n .

Common error. Coefficient matching means matching an identity for all n , not selecting a few convenient values unless those values are used in a justified polynomial-identity argument. Expanding once and comparing coefficients is usually the clearest route.

After finding the particular solution, return to the complete homogeneous-plus-particular expression before applying the initial data.

26 Exponential forcing without resonance

For forcing $c\lambda^n$, try $p_n = K\lambda^n$ if λ is not a characteristic root. Shifting this trial produces the same exponential shape, so all terms can be reduced to a coefficient equation for K .

Worked example. Solve $a_n = 2a_{n-1} + 3^n$ with $a_0 = 0$. The homogeneous root is 2, different from the forcing base 3. Try $p_n = K3^n$. Then

$$K3^n - 2K3^{n-1} = K3^{n-1}.$$

The right side must equal $3^n = 3 \cdot 3^{n-1}$, so $K = 3$. The full solution is $C2^n + 3 \cdot 3^n$. Using $a_0 = 0$ gives $C = -3$, hence $a_n = 3(3^n - 2^n)$.

The factor 3 in the particular solution is easy to miss if terms are divided by different powers. Choose one common exponential factor, cancel it consistently, and compare the remaining coefficients.

Practice. Find a particular solution of $b_n = 4b_{n-1} + 2^n$.

Solution. Try $p_n = K2^n$. Then $p_n - 4p_{n-1} = K2^n - 4K2^{n-1} = -K2^n$. Thus $K = -1$, and $p_n = -2^n$ is a particular solution. Without an initial value, the full family is $b_n = C4^n - 2^n$.

Common error. The forcing base is the number raised to n , not its coefficient. For $7 \cdot 3^n$, test whether 3 is a characteristic root. The coefficient 7 affects the equation for K but does not determine resonance.

This method also works with negative real bases, producing alternating-sign forcing, as long as the base is nonzero and the trial does not overlap the homogeneous solution.

27 Resonance: when the first trial is homogeneous

If the forcing base is already a characteristic root, a trial $K\lambda^n$ produces zero under the homogeneous recurrence operator. It cannot reproduce a nonzero forcing of that shape. This overlap is often called *resonance*.

For a simple matching root, multiply the trial by n . For multiplicity s , multiply by n^s . More generally, forcing $P_d(n)\lambda^n$, where P_d is a polynomial of degree d , calls for

$$p_n = n^s \lambda^n Q_d(n),$$

where Q_d is a general degree- d polynomial and s is the multiplicity of λ as a characteristic root, or zero if it is not a root.

Worked example. Solve $a_n = 2a_{n-1} + 2^n$ with $a_0 = 1$. The root 2 matches the forcing base. Try $p_n = Kn2^n$. Then

$$Kn2^n - 2K(n-1)2^{n-1} = K2^n,$$

so $K = 1$. The complete solution is $(C + n)2^n$. The initial value gives $C = 1$, hence $a_n = (n + 1)2^n$.

Practice. Find a particular solution to $b_n = 3b_{n-1} + 5 \cdot 3^n$.

Solution. The root 3 has multiplicity one, so try $p_n = Kn3^n$. Subtracting $3p_{n-1}$ leaves $K3^n$. Therefore $K = 5$ and $p_n = 5n3^n$. The general solution is $(C + 5n)3^n$.

Common error. Multiply the entire trial by n^s , including all lower-degree terms of its polynomial factor. An arbitrary collection of extra terms can still overlap the homogeneous family and make the coefficient equations unnecessarily singular.

Resonance is an algebraic issue here. It means that the natural response shape already solves the zero-forcing equation, so a modified shape is needed to produce the required nonzero input.

28 Repeated resonance and second differences

A double characteristic root at 1 corresponds to the second-difference operator

$$\Delta^2 a_n = a_n - 2a_{n-1} + a_{n-2}.$$

This operator annihilates constant and linear sequences. A quadratic sequence has a constant second difference, making it the natural response to constant forcing.

Worked example. Solve $a_n = 2a_{n-1} - a_{n-2} + 2$ with $a_0 = 0, a_1 = 1$. The characteristic polynomial is $(r - 1)^2$, so the homogeneous part is $A + Bn$. For the constant forcing, use $p_n = Kn^2$:

$$K(n^2 - 2(n - 1)^2 + (n - 2)^2) = 2K.$$

Set $2K = 2$, giving $K = 1$. Thus $a_n = A + Bn + n^2$. The initial conditions give $A = B = 0$, so $a_n = n^2$.

Practice. Find a particular solution to $b_n - 2b_{n-1} + b_{n-2} = 6n$.

Solution. Because the forcing is linear and 1 is a double root, try $p_n = n^2(An + B) = An^3 + Bn^2$. The second differences satisfy

$$\Delta^2(n^3) = 6n - 6, \quad \Delta^2(n^2) = 2.$$

Thus $6An - 6A + 2B = 6n$, giving $A = 1$ and $B = 3$. One particular solution is $n^3 + 3n^2$. The complete homogeneous part $C + Dn$ may be added as needed.

Common error. The second difference is centred according to the indices used here, so the second difference of n^3 is $6n - 6$, not $6n$. Expanding the shifted terms is safer than importing a formula written with a different index convention.

This section explains the multiplicity rule concretely: each difference lowers a polynomial's degree, so repeated annihilation requires additional powers of n in the response.

29 Several forcing terms and superposition

For a linear recurrence, different forcing components can be solved separately. If $L[p]_n = g_n$ and $L[q]_n = h_n$, then $L[p + q]_n = g_n + h_n$. This is the *superposition principle*.

The homogeneous solution is added only once. Separate particular solutions need not satisfy the initial data individually; the combined solution is adjusted afterward.

Worked example. Solve $a_n = 2a_{n-1} + n + 3^n$ with $a_0 = 0$. From earlier calculations, a particular solution for forcing n is $-n - 2$, and one for 3^n is $3 \cdot 3^n$. Therefore

$$a_n = C2^n - n - 2 + 3 \cdot 3^n.$$

At $n = 0$, $0 = C - 2 + 3$, giving $C = -1$. Hence $a_n = -2^n - n - 2 + 3^{n+1}$. At $n = 1$, the formula gives 4, matching $2a_0 + 1 + 3$.

Practice. State a suitable particular-solution trial for

$$b_n - 3b_{n-1} + 2b_{n-2} = n + 2^n.$$

Solution. The characteristic roots are 1 and 2, both simple. The polynomial forcing n has implicit exponential base 1, so its trial must be $n(An + B)$. The exponential forcing 2^n also resonates, giving $Cn2^n$. Combine them as

$$p_n = n(An + B) + Cn2^n.$$

The letters A, B, C are unknown coefficients to be determined by substitution. They are not yet the arbitrary constants of the homogeneous solution.

Common error. Resonance must be checked separately for each forcing component. A recurrence can resonate with one component but not another, or with different multiplicities for different bases.

Superposition applies because the recurrence is linear. For a nonlinear update such as $a_n = (a_{n-1})^2 + g_n$, adding solutions generally does not add their forcing terms in the same way.

30 A complete recurrence-solving workflow

For a linear recurrence with constant coefficients, first identify the order, the starting index, the initial values, and the forcing. Put the recurrence into a consistent left-side form. Solve its homogeneous characteristic equation, respecting multiplicities. Then choose and verify a particular solution, combine the parts, and fit the initial conditions.

Worked example. Solve $a_n = 3a_{n-1} - 2a_{n-2} + 2^n$ for $n \geq 2$, with $a_0 = 0, a_1 = 0$. The homogeneous roots are 1, 2, so $h_n = A + B2^n$. Since 2 is a simple root, try $p_n = Cn2^n$. Substitution gives

$$p_n - 3p_{n-1} + 2p_{n-2} = \frac{C}{2} 2^n.$$

Thus $C = 2$, and $a_n = A + B2^n + 2n2^n$. The initial equations are $A + B = 0$ and $A + 2B + 4 = 0$. Therefore $B = -4, A = 4$, giving

$$a_n = 4 + (2n - 4)2^n.$$

At $n = 2$, this gives 4, agreeing with the recurrence's $0 - 0 + 4$.

Practice. Where would the calculation change if the forcing were 5^n instead?

Solution. The homogeneous roots and initial-condition count remain the same. Because 5 is not a root, the particular trial becomes $C5^n$ with no factor n . Substitution gives $C5^n(1 - 3/5 + 2/25) = \frac{12C}{25}5^n$, so $C = 25/12$. New initial equations then determine the homogeneous constants.

Common error. A spot check at one new index is useful for detecting errors but is not full verification. The particular-solution substitution and characteristic equation together establish the recurrence symbolically; the final initial checks identify the intended sequence.

When a trial becomes lengthy, keep the homogeneous and particular calculations visually separate. This makes it easier to locate whether an error comes from factoring, forcing coefficients, or starting data.

31 Application: Fibonacci numbers and tilings

The Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Its characteristic equation is $r^2 - r - 1 = 0$, with roots

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

Therefore $F_n = A\phi^n + B\psi^n$. The initial conditions give $A + B = 0$ and $A(\phi - \psi) = 1$. Since $\phi - \psi = \sqrt{5}$,

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}.$$

Although this formula uses irrational numbers, the recurrence and initial values show that its values are integers.

Worked example. The strip-tiling sequence from Section 14 has $t_0 = t_1 = 1$ and the same recurrence. Its starting values match F_1, F_2 , so $t_n = F_{n+1}$. In particular, a strip of length five has $F_6 = 8$ tilings.

Practice. How many ways can a person climb a staircase of six steps using moves of one or two steps? Explain why the same recurrence applies.

Solution. Let s_n count step sequences that climb n steps. The last move has size one or two, leaving s_{n-1} or s_{n-2} possibilities. The empty climb gives $s_0 = 1$, and $s_1 = 1$. Hence $s_n = F_{n+1}$ and $s_6 = F_7 = 13$.

Common error. A common recurrence does not imply identical sequences unless the initial values also match. Fibonacci numbers, strip tilings, and staircase counts differ by an index shift under these conventions.

When interpreting a formula, connect its indices back to the original objects. The subscript counts steps or strip length, not the number of pieces in a particular arrangement.

32 Application: binary strings with restrictions

Let b_n count length- n binary strings with no adjacent 1s. The empty string is allowed, so $b_0 = 1$, and the length-one strings are 0, 1, giving $b_1 = 2$.

For $n \geq 2$, a valid string either ends in 0 or in 1. If it ends in 0, removing that last symbol leaves any valid length- $(n - 1)$ string. If it ends in 1, the preceding symbol must be 0; removing the final 01 leaves any valid length- $(n - 2)$ string. Thus

$$b_n = b_{n-1} + b_{n-2}, \quad b_n = F_{n+2}.$$

The recurrence arises from the restriction, not from a numerical coincidence in the first few counts.

Worked example. For length three, the valid strings are 000, 001, 010, 100, 101. Thus $b_3 = 5$, agreeing with $F_5 = 5$. The prohibited strings 011, 110, 111 each contain adjacent 1s.

Practice. Find the number of valid strings of length five. Then explain how the recurrence changes if all binary strings are permitted.

Solution. The restricted values are 1, 2, 3, 5, 8, 13, so $b_5 = 13$. Without the restriction, either final symbol can follow any earlier string. If u_n counts all strings, $u_n = 2u_{n-1}$ with $u_0 = 1$, giving $u_n = 2^n$. The restriction changes the size of the earlier object left after removing a final 1.

Common error. When removing a suffix, prove the reverse construction also works. Here appending 01 to any valid shorter string introduces no adjacent 1s because the appended 0 separates the old end from the new 1.

More complicated forbidden patterns may require several state counts, such as counts indexed by the current suffix. A single total may not contain enough information to describe the next extension correctly.

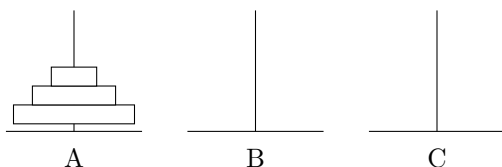
33 Application: the Tower of Hanoi

In the three-peg Tower of Hanoi, n disks of different sizes begin stacked on one peg. A move transfers one top disk, and a larger disk may never rest on a smaller one. Let H_n be the minimum moves needed to transfer the stack to another peg.

To move the largest disk, all $n - 1$ smaller disks must first be on the third peg. This requires at least H_{n-1} moves. Moving the largest disk takes one move. Transferring the smaller stack onto it takes at least another H_{n-1} . The standard recursive procedure achieves exactly these stages, so

$$H_n = 2H_{n-1} + 1, \quad H_0 = 0.$$

Repeated substitution or a constant particular solution gives $H_n = 2^n - 1$.



Practice. Find H_4 and explain why the recurrence establishes a minimum rather than only the cost of one algorithm.

Solution. $H_4 = 2^4 - 1 = 15$. The argument contains a lower bound: any legal solution must clear the largest disk, move it, and then place the smaller disks above it. It also contains a construction achieving that bound. A cost calculation without the necessity argument would establish only an upper bound on the minimum.

Common error. The recurrence relies on exactly three pegs. Adding another peg changes the available strategies, so the same minimum-move proof no longer applies unchanged.

The exponential formula explains the rapid growth: adding one disk doubles the previous task and adds one unavoidable move.

34 Application: regions formed by lines

Let R_n be the largest number of regions into which n lines divide the plane. To achieve the maximum, assume no two lines are parallel and no three pass through the same point.

When the n th line is added, it intersects the previous $n - 1$ lines at distinct points. These points divide the new line into n segments, counting the two unbounded end pieces. Each segment passes through and splits one existing region, so the new line adds n regions. Hence

$$R_n = R_{n-1} + n, \quad R_0 = 1.$$

Summing the increments gives

$$R_n = 1 + \sum_{j=1}^n j = 1 + \frac{n(n+1)}{2}.$$

Worked example. For three lines in general position, $R_3 = 7$. If the three lines meet at one common point, there are only six regions, so the general-position assumptions are doing real work.

Practice. What is the maximum for five lines? What is the region count if all five lines are parallel?

Solution. The maximum is $R_5 = 1 + 5 \cdot 6/2 = 16$. Five distinct parallel lines create six regions, because each added line contributes only one new region. The recurrence for that restricted arrangement is $P_n = P_{n-1} + 1$, with $P_0 = 1$.

Common error. A maximum-count recurrence needs both a limit on how many new regions can be created and an arrangement achieving that limit. The new line has at most $n - 1$ distinct intersections, so it creates at most n new regions; general position achieves this number.

This is an additive recurrence even though the drawing becomes visually complicated. Counting the *change* produced by one new object is often easier than counting the entire resulting arrangement directly.

35 Application: accumulated resources and model boundaries

A recurrence can model an evolving quantity when each step retains a fixed fraction of the previous amount and receives a new input. Let a_n be the amount after step n , r the retained multiplier, and c the input added at the end of each step. Then

$$a_n = ra_{n-1} + c.$$

This is a mathematical model; its interpretation depends on the order of retention and input. If the input arrives before multiplication, the rule becomes $a_n = r(a_{n-1} + c)$ instead.

Worked example. A container retains half its contents between measurements, then receives three units. With $a_0 = 10$, the recurrence is $a_n = \frac{1}{2}a_{n-1} + 3$. The fixed value satisfies $L = \frac{1}{2}L + 3$, giving $L = 6$. Subtracting this value yields

$$a_n - 6 = \frac{1}{2}(a_{n-1} - 6),$$

so $a_n = 6 + 4(1/2)^n$. The values approach 6 because the difference from 6 halves each step.

Practice. If three units are added before half the contents are lost, what is the recurrence and its fixed value?

Solution. The recurrence is $b_n = \frac{1}{2}(b_{n-1} + 3) = \frac{1}{2}b_{n-1} + \frac{3}{2}$. A fixed value satisfies $L = \frac{1}{2}L + \frac{3}{2}$, giving $L = 3$. Changing the timing changes the eventual amount, even though the same numbers appear in the verbal description.

Common error. Do not use a constant-coefficient solution method automatically for rules like $a_n = na_{n-1} + 1$ or $a_n = a_{\lfloor n/2 \rfloor} + n$. Those are different recurrence classes. They may be solvable, but their structure does not match the characteristic-polynomial method developed here.

Defining the measured quantity and update timing precisely is part of solving an application, not a preliminary formality.

36 Ordinary generating functions: the basic idea

The ordinary generating function of a sequence $(a_n)_{n \geq 0}$ is

$$A(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots.$$

The variable x is a bookkeeping device. Its exponent records the index, and the coefficient records the sequence value. We write $[x^n]A(x) = a_n$ to mean “the coefficient of x^n in $A(x)$.”

Main idea. A generating function packages infinitely many related equations into one algebraic expression. Index shifts become multiplication by powers of x , and combining independent choices becomes multiplication of series.

Worked example. The sequence $1, 1, 1, \dots$ has generating function $1 + x + x^2 + \cdots = 1/(1 - x)$. The sequence $1, 2, 4, 8, \dots$ has generating function $1/(1 - 2x)$. The second formula follows by substituting $2x$ into the geometric-series identity.

In these notes, series are primarily *formal power series*. Equality means equality of every coefficient; numerical convergence is not needed for the algebraic counting arguments. If x is instead given a numerical value, convergence must be checked separately. For example, the geometric series converges numerically only when the magnitude of its ratio is less than 1.

Practice. What are $[x^0](3 + 5x^2 + 7x^4)$ and $[x^3](3 + 5x^2 + 7x^4)$? Which sequence does the polynomial encode?

Solution. The coefficients are 3 and 0. The encoded sequence is $3, 0, 5, 0, 7, 0, 0, \dots$. Missing powers have zero coefficients. The coefficient of x^0 is the constant term, so the first sequence value is not attached to x^1 .

Common error. Do not confuse $A(3)$, evaluation at the number 3, with $[x^3]A(x)$, extraction of the coefficient at index three. They are entirely different operations.

37 A small library of generating functions

Several basic coefficient patterns are worth understanding well:

$$\begin{aligned}\frac{1}{1-x} &= \sum_{n \geq 0} x^n, \\ \frac{1}{1-rx} &= \sum_{n \geq 0} r^n x^n, \\ \frac{1}{(1-x)^2} &= \sum_{n \geq 0} (n+1)x^n, \\ \frac{x}{(1-x)^2} &= \sum_{n \geq 0} nx^n.\end{aligned}$$

The third identity follows by multiplying two copies of $1/(1-x)$: the coefficient of x^n counts pairs of nonnegative integers whose sum is n , and there are $n+1$ such pairs.

More generally, for an integer $m \geq 1$,

$$\frac{1}{(1-x)^m} = \sum_{n \geq 0} \binom{n+m-1}{m-1} x^n.$$

The coefficient counts ways to split n indistinguishable units among m labelled positions. It will be used for repeated poles and unrestricted counts.

Worked example. The sequence $1, 3, 5, 7, \dots$ has $a_n = 2n + 1$. By adding the corresponding series, its generating function is

$$\frac{2x}{(1-x)^2} + \frac{1}{1-x} = \frac{1+x}{(1-x)^2}.$$

Practice. Find the generating function of $a_n = 4 \cdot 3^n + 2$ for $n \geq 0$.

Solution. Linearity gives $A(x) = 4/(1-3x) + 2/(1-x)$. At x^0 , the coefficient is $4 + 2 = 6$, matching a_0 . At x^1 , it is $12 + 2 = 14$, matching a_1 . These initial coefficients are convenient checks on the construction.

Common error. The coefficient of x^n in $(1-x)^{-2}$ is $n+1$, not n . Multiplication by x shifts the coefficients and produces the series for n .

When a standard identity is forgotten, recover it from the geometric series, a product, or a formal derivative instead of relying on a possibly misremembered index.

38 Shifting sequences and correcting initial terms

Multiplying by x shifts a sequence one place to the right:

$$xA(x) = 0 + a_0x + a_1x^2 + \cdots.$$

Removing the first term and shifting left gives

$$\frac{A(x) - a_0}{x} = a_1 + a_2x + a_3x^2 + \cdots.$$

These formal quotients are valid because the numerator has zero constant term and is divisible by x .

When summing a recurrence beginning at $n = 2$, the useful identities are

$$\begin{aligned}\sum_{n \geq 2} a_n x^n &= A - a_0 - a_1 x, \\ \sum_{n \geq 2} a_{n-1} x^n &= x(A - a_0), \\ \sum_{n \geq 2} a_{n-2} x^n &= x^2 A.\end{aligned}$$

The subtracted terms record indices excluded from the summation. They are where initial conditions enter the generating-function equation.

Worked example. If $A(x) = 1 + 3x + 5x^2 + \cdots$, then $x(A(x) - 1) = 3x^2 + 5x^3 + \cdots$. This is exactly the series $\sum_{n \geq 2} a_{n-1} x^n$, whose first term uses a_1 rather than a_0 .

Practice. Express $\sum_{n \geq 3} a_{n-2} x^n$ in terms of $A(x)$ and the initial coefficients.

Solution. Put $j = n - 2$. Then $j \geq 1$ and $x^n = x^{j+2}$. Hence the sum is $x^2 \sum_{j \geq 1} a_j x^j = x^2(A(x) - a_0)$. The subtraction is required because the original sum begins at $n = 3$, not $n = 2$.

Common error. A shift changes both the exponent and the lower summation limit. Write the first two terms of the shifted sum whenever the correction is uncertain. This simple check prevents most generating-function recurrence errors.

Treat A without its argument as an abbreviation for $A(x)$ only after defining it. The shorter notation can simplify equations without changing their meaning.

39 Products and convolution

If $A(x) = \sum a_n x^n$ and $B(x) = \sum b_n x^n$, then

$$[x^n]A(x)B(x) = \sum_{j=0}^n a_j b_{n-j}.$$

This coefficient sum is called a *convolution*. To make exponent n in the product, choose exponent j from the first series and exponent $n - j$ from the second, then add over every possible split.

Worked example. Multiply $(1 + x + x^2)$ by $(1 + x)$. The coefficient of x^2 is $1 \cdot 1 + 1 \cdot 1 = 2$, arising from $x^2 \cdot 1$ and $x \cdot x$. The full product is $1 + 2x + 2x^2 + x^3$.

In a counting interpretation, the product counts pairs of independently selected component objects, with total size equal to the sum of their sizes. The coefficients need not be 0 or 1: a coefficient greater than one means several components have that same size.

Practice. Find $[x^3](1 + x + x^2 + x^3)^2$ and explain it combinatorially.

Solution. The contributing exponent pairs are $(0, 3), (1, 2), (2, 1), (3, 0)$. Each contributes one, so the coefficient is four. Equivalently, it counts ordered pairs (u, v) with $0 \leq u, v \leq 3$ and $u + v = 3$. The positions are labelled by the two factors, so $(0, 3)$ and $(3, 0)$ are different outcomes.

Common error. Multiplying generating functions does not multiply corresponding coefficients term by term. The coefficient of x^n is a sum over all index splits, not merely $a_n b_n$.

Formal multiplication is well defined because any fixed coefficient uses only finitely many terms, even when both series are infinite. This is one reason generating functions can be manipulated without treating x as a numerical value.

40 Counting with coin-type generating functions

Suppose an item contributes size d and may be chosen any nonnegative number of times. Its possible contributions are represented by

$$1 + x^d + x^{2d} + \cdots = \frac{1}{1 - x^d}.$$

The exponent records total size, while the selected term records the multiplicity of that item. Multiplying one factor for each labelled item type counts combinations of multiplicities.

Worked example. With denominations 1, 2, 5 and unlimited supplies, the number of combinations totalling n is

$$[x^n] \frac{1}{(1 - x)(1 - x^2)(1 - x^5)}.$$

For $n = 5$, either one 5 is used, giving one combination, or no 5 is used. In the latter case, the number of 2s can be 0, 1, 2, and the remaining amount is filled with 1s. Thus the coefficient is four.

Practice. Count the combinations totalling 6 using unlimited 1, 2, and 3 units.

Solution. The generating function is $1/((1 - x)(1 - x^2)(1 - x^3))$. Condition on the number c of 3-unit items. If $c = 0$, the number of 2s can be 0, 1, 2, 3, giving four choices. If $c = 1$, the remaining total is 3, giving two choices for the number of 2s. If $c = 2$, only one combination remains. The total is $4 + 2 + 1 = 7$.

Common error. This product counts combinations, not ordered sequences of coin placements. A 1 followed by a 2 and a 2 followed by a 1 have the same multiplicities and therefore contribute one combination, not two.

If order matters, a different construction is needed. For sequences of parts of sizes 1 and 2, the generating function is $1/(1 - x - x^2)$, because each new part may be chosen in either size. The distinction between a product over types and a sequence of choices is essential.

41 Bounded selections and minimum requirements

Generating functions encode restrictions directly in their factors. If a type of size d may occur at most b times, use $1 + x^d + \cdots + x^{bd}$. If it must occur at least a times with no upper bound, use $x^{ad}/(1 - x^d)$.

Worked example. Count integer solutions of $u + v + w = 6$ with $0 \leq u \leq 2$, $v \geq 1$, and $w \geq 0$. The generating function is

$$(1 + x + x^2) \frac{x}{1 - x} \frac{1}{1 - x}.$$

The desired coefficient is $[x^5](1 + x + x^2)/(1 - x)^2$. Since $[x^m](1 - x)^{-2} = m + 1$, it equals $6 + 5 + 4 = 15$. These three contributions correspond to $u = 0, 1, 2$.

Practice. How many solutions satisfy $u + v + w = 5$ with $0 \leq u, v \leq 2$ and $w \geq 0$?

Solution. Use $(1 + x + x^2)^2/(1 - x)$. Every one of the nine ordered pairs (u, v) has $u + v \leq 4$, so exactly one nonnegative $w = 5 - u - v$ completes it. Hence the coefficient is nine. The generating function records the same reasoning: summing the coefficients of the bounded product through degree five includes all its nine choices.

Common error. A lower bound is implemented by a shift, not by deleting an arbitrary number of coefficients from the final product. For $v \geq 1$, the factor begins with x , whereas an unrestricted variable begins with 1.

For a finite geometric factor, the identity $1 + x + \cdots + x^b = (1 - x^{b+1})/(1 - x)$ can make coefficient extraction easier. The numerator subtracts precisely those choices that exceed the bound, connecting generating functions with inclusion–exclusion.

Always define whether the variables are labelled. In these equations, (u, v, w) is ordered because the three positions have distinct roles.

42 Solving a first-order recurrence by generating functions

Generating functions solve recurrences by turning index shifts into algebraic factors. Define the series, multiply the recurrence by x^n , sum over exactly the valid indices, and substitute the shift identities before rearranging.

Worked example. Solve $a_n = 2a_{n-1} + 1$ for $n \geq 1$, with $a_0 = 0$. Let $A(x) = \sum_{n \geq 0} a_n x^n$. Summing gives

$$A(x) = 2xA(x) + \frac{x}{1-x}.$$

The forcing sum begins at $n = 1$, so it is $x/(1-x)$ rather than $1/(1-x)$. Rearranging gives

$$A(x) = \frac{x}{(1-x)(1-2x)} = \frac{1}{1-2x} - \frac{1}{1-x}.$$

Extracting coefficients yields $a_n = 2^n - 1$.

Practice. Derive a generating function for $b_n = 3b_{n-1}$, $n \geq 1$, with $b_0 = 2$, and recover its formula.

Solution. Summation gives $B(x) - 2 = 3xB(x)$. Hence $B(x) = 2/(1-3x)$. The coefficient of x^n is $2 \cdot 3^n$, so $b_n = 2 \cdot 3^n$. The subtraction of 2 on the left accounts for the initial term not governed by the recurrence.

Common error. Do not sum the recurrence from $n = 0$ when it starts at 1. Doing so introduces an undefined a_{-1} and usually loses the initial condition. The lower limit is part of the mathematics, not a cosmetic choice.

The rational function and the closed formula contain the same sequence information in different forms. The rational form is often more convenient for algebra, while the coefficient formula is more convenient for computing a distant individual term.

43 Second-order recurrences and their generating functions

For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ with initial values a_0, a_1 , sum from $n = 2$:

$$A - a_0 - a_1 x = c_1 x(A - a_0) + c_2 x^2 A.$$

Rearranging gives the general formula

$$A(x) = \frac{a_0 + (a_1 - c_1 a_0)x}{1 - c_1 x - c_2 x^2}.$$

The denominator records the recurrence coefficients, and the numerator records the initial conditions. Both are necessary to specify the sequence.

Worked example. For Fibonacci numbers, $a_0 = 0, a_1 = 1, c_1 = c_2 = 1$, so

$$F(x) = \frac{x}{1 - x - x^2}.$$

For the no-adjacent-1s count with $b_0 = 1, b_1 = 2$, the same denominator occurs but the numerator becomes $1 + x$. Thus $B(x) = (1 + x)/(1 - x - x^2)$, reflecting the different starting values.

Practice. Find the generating function of $a_n = 3a_{n-1} - 2a_{n-2}$ with $a_0 = 0, a_1 = 1$.

Solution. The formula gives

$$A(x) = \frac{x}{1 - 3x + 2x^2} = \frac{x}{(1 - x)(1 - 2x)}.$$

As in the previous section, partial fractions yield $A(x) = 1/(1 - 2x) - 1/(1 - x)$, so $a_n = 2^n - 1$. The generating-function result agrees with the characteristic-root calculation.

Common error. The denominator uses the signs from moving recurrence terms to the left. In particular, a recurrence coefficient $c_2 = -2$ produces $+2x^2$ in the denominator.

You do not need to memorise the general formula if the shift identities are secure. Deriving it takes only a few lines and reduces the chance of using the wrong starting-index convention.

44 Partial fractions and coefficient extraction

Partial fractions split a rational generating function into standard pieces whose coefficients are known. When the denominator factors into distinct linear terms, try one constant numerator over each factor.

Worked example. Find the sequence encoded by

$$A(x) = \frac{1 + 2x}{(1 - x)(1 - 3x)}.$$

Write $A(x) = C/(1 - x) + D/(1 - 3x)$. Multiplying by the denominator gives

$$1 + 2x = C(1 - 3x) + D(1 - x).$$

Matching coefficients yields $C + D = 1$ and $-3C - D = 2$. Thus $C = -3/2$, $D = 5/2$, and

$$a_n = -\frac{3}{2} + \frac{5}{2} 3^n.$$

Although fractions appear in the formula, the coefficient at $n = 0$ is 1, and later values are integers because 3^n is odd.

Practice. Extract $[x^n]1/((1 - x)(1 - 2x))$.

Solution. Write the fraction as $C/(1 - x) + D/(1 - 2x)$. The equations $C + D = 1$ and $2C + D = 0$ give $C = -1$, $D = 2$. Thus the coefficient is $-1 + 2 \cdot 2^n = 2^{n+1} - 1$. A check at $n = 0$ gives 1, matching the constant term of the original product.

Common error. If the numerator degree is at least the denominator degree, perform polynomial division first. The polynomial part contributes finitely many initial coefficients and should not be discarded.

When the desired index is small, direct multiplication can be easier than partial fractions. Use algebraic decomposition when you need a general coefficient formula or a distant term, and choose the amount of calculation appropriate to the question.

45 Repeated factors and polynomial coefficient growth

A repeated factor requires every denominator power up to its multiplicity in a partial-fraction decomposition. For example, a factor $(1 - rx)^2$ contributes terms of the forms $C/(1 - rx)$ and $D/(1 - rx)^2$.

The coefficient identity is

$$[x^n] \frac{1}{(1 - rx)^m} = \binom{n + m - 1}{m - 1} r^n \quad (m \geq 1).$$

For $m = 2$, this is $(n + 1)r^n$. The repeated denominator creates a polynomial factor in n , matching the repeated-root solution form for recurrences.

Worked example. The sequence $a_n = (n + 1)2^n$ has generating function $1/(1 - 2x)^2$. The sequence $n2^n$ has generating function $2x/(1 - 2x)^2$, obtained by shifting the coefficients appropriately or subtracting $1/(1 - 2x)$ from the first series.

Practice. Find the coefficient of x^n in $(1 + x)/(1 - x)^3$.

Solution. The unshifted part contributes $\binom{n+2}{2}$. For $n \geq 1$, the shifted part contributes $\binom{n+1}{2}$. Their sum is

$$\frac{(n + 2)(n + 1) + n(n + 1)}{2} = (n + 1)^2.$$

At $n = 0$, only the first part contributes, giving 1, which also equals $(n + 1)^2$. Thus the answer is valid for every $n \geq 0$.

Common error. A factor x shifts the index of the entire coefficient formula. The coefficient of x^n in $x/(1 - x)^3$ is based on $n - 1$, not n . Check the constant term whenever a shifted series appears.

Repeated-factor decomposition is the generating-function version of repeated characteristic roots. Learning the connection reduces the number of separate patterns that need to be remembered.

46 Nonhomogeneous recurrences by generating functions

The generating-function method handles forcing by inserting its own series. This often avoids choosing a particular-solution trial explicitly. The same requirements remain: correct starting indices and accurate initial-term corrections.

Worked example. Solve $a_n = 2a_{n-1} + n$ with $a_0 = 0$. Let $A(x) = \sum a_n x^n$. Since $\sum_{n \geq 1} n x^n = x/(1-x)^2$, summing yields

$$A(x) = 2xA(x) + \frac{x}{(1-x)^2}.$$

Therefore

$$\begin{aligned} A(x) &= \frac{x}{(1-2x)(1-x)^2} \\ &= \frac{2}{1-2x} - \frac{1}{1-x} - \frac{1}{(1-x)^2}. \end{aligned}$$

Coefficient extraction gives $a_n = 2^{n+1} - 1 - (n+1) = 2^{n+1} - n - 2$, agreeing with the earlier method.

Practice. Use generating functions to solve $b_n = b_{n-1} + 1$ with $b_0 = 0$.

Solution. Summation gives $B = xB + x/(1-x)$, so $B = x/(1-x)^2$. Its coefficient is $b_n = n$. The two copies of $1-x$ have different origins: one comes from the update rule, and one from the constant forcing. Their repeated appearance is the generating-function form of resonance at root 1.

Common error. For forcing g_n summed only from $n = 2$, use $G(x) - g_0 - g_1x$ if G was defined from $n = 0$. The forcing requires initial-term corrections just as the sequence itself does.

The algebra also explains why the answer may contain both exponential and polynomial terms: the final denominator can contain both $1-rx$ from the homogeneous recurrence and powers of $1-x$ from polynomial forcing.

47 Building generating functions from object constructions

Generating functions can be derived directly from how objects are built. If a single component has generating function $C(x)$ with zero constant term, then an ordered sequence of zero or more such components has generating function

$$1 + C(x) + C(x)^2 + \cdots = \frac{1}{1 - C(x)}.$$

The zero constant term ensures that a component has positive size, so only finitely many component counts contribute to any fixed total size.

Worked example. For tilings by pieces of lengths one and two, one piece contributes $C(x) = x + x^2$. A tiling is an ordered sequence of pieces, so

$$T(x) = \frac{1}{1 - x - x^2}.$$

This agrees with the recurrence and $t_0 = t_1 = 1$. The constant term 1 represents the empty tiling.

Practice. Find a generating function for ordered compositions of an integer using only parts 1 and 3. Determine the coefficient at x^4 .

Solution. One part contributes $x + x^3$, so the sequence construction gives $1/(1 - x - x^3)$. For total four, the possible part sequences are $(1, 1, 1, 1)$, $(1, 3)$, and $(3, 1)$, giving coefficient three. If c_n denotes the count, then $c_n = c_{n-1} + c_{n-3}$ with $c_0 = 1$ and $c_n = 0$ for negative indices provides a uniform recurrence for $n \geq 1$.

Common error. In this construction, powers of $C(x)$ count ordered lists of components. Replacing it by a product such as $1/((1 - x)(1 - x^3))$ would count unordered multiplicities of the two sizes and would merge $(1, 3)$ with $(3, 1)$.

A generating-function equation should have a combinatorial explanation for each sum and product. This keeps the algebra grounded in what the coefficients actually count.

48 Cumulative sums, differentiation, and useful operations

Multiplying by $1/(1-x)$ forms cumulative sums. If $b_n = a_0 + \cdots + a_n$, then convolution gives

$$B(x) = \frac{A(x)}{1-x}.$$

Taking a formal derivative weights coefficients by their indices:

$$\begin{aligned} A'(x) &= \sum_{n \geq 1} n a_n x^{n-1}, \\ xA'(x) &= \sum_{n \geq 0} n a_n x^n. \end{aligned}$$

The prime denotes differentiation. In formal-series work, this is defined coefficient by coefficient and does not require an analytic convergence argument.

Worked example. Starting with $A(x) = 1/(1-x)$, differentiation gives $1/(1-x)^2$, and multiplication by x gives the series for n . Applying $x d/dx$ once more gives

$$\sum_{n \geq 0} n^2 x^n = \frac{x(1+x)}{(1-x)^3}.$$

This identity is useful when a recurrence has quadratic forcing.

Practice. Find the generating function of $b_n = 1 + 2 + \cdots + (n+1)$ for $n \geq 0$, and extract its formula.

Solution. The sequence being cumulatively summed is $a_n = n+1$, whose generating function is $(1-x)^{-2}$. Therefore $B(x) = (1-x)^{-3}$. Its coefficient is $\binom{n+2}{2} = (n+1)(n+2)/2$, as expected from the arithmetic-sum formula.

Common error. Differentiation alone shifts exponents downward. Multiplication by x restores the original index positions. Omitting this factor gives the series for $(n+1)a_{n+1}$ instead of na_n .

Operations on sequences become operations on their generating functions: addition is termwise addition, shifting is multiplication by x , cumulative summation is division by $1-x$, and index weighting is x times differentiation. Keeping this small dictionary available makes unfamiliar examples more approachable.

49 Connecting roots, denominators, and growth

For a homogeneous recurrence with characteristic roots r_1, r_2 , the generating-function denominator factors as $(1 - r_1x)(1 - r_2x)$. The root r in the recurrence corresponds to the denominator factor $1 - rx$, whose coefficients grow like r^n .

A repeated factor of multiplicity m produces a polynomial of degree at most $m - 1$ multiplying r^n . This is the same pattern obtained from repeated characteristic roots. The two methods are different representations of the same linear structure.

Worked example. If $a_n = 2^n + 3^n$, then $A(x) = 1/(1 - 2x) + 1/(1 - 3x)$. Dividing by 3^n gives $a_n/3^n = (2/3)^n + 1$, which approaches 1. Thus the 3^n component determines the eventual scale.

For Fibonacci numbers, $|\psi| < 1$ while $\phi > 1$. Therefore the correction $\psi^n/\sqrt{5}$ becomes small in magnitude, and F_n is close to $\phi^n/\sqrt{5}$ for large n . This is an approximation to the exact two-root formula, not a replacement for it at every stage of an exact proof.

Practice. A recurrence permits solutions $a_n = A2^n + B5^n$. Must every solution grow like 5^n ?

Solution. No. If the initial conditions make $B = 0$, the 5^n component is absent. For example, $a_n = 2^n$ is a valid solution of that recurrence family. In the generating function, the unused factor can cancel against the numerator. Growth depends on the roots that actually remain with nonzero coefficients.

Common error. A denominator factor before simplification does not always contribute to the final sequence. Check cancellation and coefficients before identifying a dominant term. Equal magnitudes with different signs or complex phases can also produce oscillation and cancellation.

For exact counting questions, retain the exact formula or recurrence. Growth comparisons explain behaviour, but a rounded approximation may not supply the requested integer count.

50 Revision guide and notation reference

The three main ideas in the unit answer different questions. Pigeonhole reasoning forces existence from limited categories. Recurrences describe how quantities change from one index to the next. Generating functions encode all sequence values in one algebraic object and convert shifts or combinations into manageable operations.

Notation	Meaning
$ S $	number of elements of a finite set
$\lceil t \rceil, \lfloor t \rfloor$	ceiling and floor
$a \equiv b \pmod{m}$	m divides $a - b$
a_n	sequence value at index n
r	a characteristic root or multiplier
h_n, p_n	homogeneous and particular parts
$\sum_{j=0}^n u_j$	sum of the indicated indexed terms
$A(x) = \sum_{n \geq 0} a_n x^n$	ordinary generating function
$[x^n]A(x)$	coefficient of x^n
$\binom{n}{r}$	number of r -element subsets of n objects

Practice. Choose a method for each task: (a) prove a repeated remainder exists; (b) solve $a_n = 5a_{n-1}$ with $a_0 = 2$; (c) count bounded solutions of an integer-sum equation; (d) explain how a no-adjacent-symbol restriction changes a string count.

Solution. (a) Use remainder classes as pigeonholes. (b) Use the geometric formula $a_n = 2 \cdot 5^n$. (c) Use one finite or shifted generating-function factor for each variable, then extract the target coefficient. (d) Classify strings by an appropriate final symbol or suffix to derive a recurrence, checking reversibility of the construction.

For a recurrence solution, verify the initial data and the update rule. For a counting model, check that cases are exhaustive and nonoverlapping. For a generating-function calculation, check the first few coefficients and every index shift. For a minimal guarantee, provide both the universal bound and an arrangement showing sharpness.

These checks are not separate from understanding. They test whether the mathematical representation still matches the original question.

51 Cumulative practice questions

Attempt these questions before reading the next section. Include the model, method, and verification, rather than only the final number or formula.

1. What is the smallest number of objects that guarantees seven in one of nine boxes? Justify minimality.
2. Prove that among any eight integers, two have a difference divisible by seven.
3. Given five integers, prove that some nonempty consecutive block has sum divisible by five.
4. Solve $a_n = 4a_{n-1} - 3a_{n-2}$ with $a_0 = 2, a_1 = 4$.
5. Solve $b_n = 6b_{n-1} - 9b_{n-2}$ with $b_0 = 1, b_1 = 6$.
6. Solve $c_n = 2c_{n-1} + 2^n$ with $c_0 = 0$.
7. Solve $d_n = d_{n-1} + 3n + 1$ with $d_0 = 2$.
8. Derive a recurrence and initial values for length- n binary strings with no adjacent 0s. Find the count at $n = 4$.
9. Find $[x^n](1+x)/(1-x)^2$.
10. How many nonnegative integer solutions satisfy $u + v + w = 8$ and $u \leq 2$? Give a generating-function solution.
11. Find the generating function of $e_n = 3e_{n-1} - 2e_{n-2}$ with $e_0 = 1, e_1 = 4$, then extract e_n .
12. Find the coefficient of x^5 in $1/(1-x-x^2)$ and explain its tiling meaning.

Working guidance. In Questions 1–3, identify the counted objects and their boxes explicitly. In Questions 4–7, distinguish the homogeneous family from a particular response and keep the starting index visible. In Question 8, justify how deleting a final symbol or suffix maps to smaller valid strings. In Questions 9–12, state which coefficient identity or shift you use.

Extension. For Question 10, solve the problem again by summing separately over $u = 0, 1, 2$. Agreement between the direct count and generating function provides a useful independent check.

Do not confuse a numerical check with a proof. Testing the first three terms helps catch algebraic mistakes, while satisfying the recurrence symbolically and matching all initial values proves the closed formula.

52 Cumulative solutions and further reading

Solutions 1–6

1. The largest avoiding arrangement has six objects in each of nine boxes, for 54 total. Therefore 55 guarantee seven. The occupancy vector $(6, 6, \dots, 6)$ shows that 54 do not suffice.
2. The eight integers occupy seven remainder classes modulo 7. Two have the same remainder, so their difference is divisible by 7.
3. Form six partial sums $S_0 = 0, S_1, \dots, S_5$. Two have the same remainder modulo 5. If their indices are $i < j$, the nonempty block from a_{i+1} through a_j sums to $S_j - S_i$, a multiple of 5.
4. The characteristic polynomial is $(r - 1)(r - 3)$. Write $a_n = A + B3^n$. The equations $A + B = 2$ and $A + 3B = 4$ give $A = B = 1$. Thus $a_n = 1 + 3^n$.
5. The polynomial is $(r - 3)^2$, so $b_n = (A + Bn)3^n$. At $n = 0$, $A = 1$. At $n = 1$, $3(1 + B) = 6$, giving $B = 1$. Hence $b_n = (n + 1)3^n$.
6. The forcing resonates with the root 2. A particular solution is $n2^n$, and the homogeneous part is $C2^n$. Since $c_0 = 0$, $C = 0$, so $c_n = n2^n$. Direct substitution gives $2(n - 1)2^{n-1} + 2^n = n2^n$.

Solutions 7–12

7. Sum the increments:

$$\begin{aligned} d_n &= 2 + \sum_{j=1}^n (3j + 1) \\ &= 2 + \frac{3n(n+1)}{2} + n \\ &= \frac{3n^2 + 5n + 4}{2}. \end{aligned}$$

The formula gives 2 at $n = 0$ and successive difference $3n + 1$.

8. Let $s_0 = 1, s_1 = 2$. For $n \geq 2$, strings ending in 1 correspond to valid length- $(n - 1)$ strings; strings ending in 0 must end in 10 and correspond to length- $(n - 2)$ strings. Thus $s_n = s_{n-1} + s_{n-2}$, giving $s_2 = 3, s_3 = 5, s_4 = 8$.
9. The unshifted part contributes $n + 1$. The shifted part contributes n for $n \geq 1$ and zero at $n = 0$. Therefore the coefficient is $2n + 1$ for every $n \geq 0$.
10. Use $(1 + x + x^2)/(1 - x)^2$. The coefficient of x^8 is $9 + 8 + 7 = 24$. Directly, fixing $u = 0, 1, 2$ leaves $v + w = 8, 7, 6$, with respectively 9, 8, 7 nonnegative solutions.
11. The second-order formula gives

$$E(x) = \frac{1 + (4 - 3)x}{1 - 3x + 2x^2} = \frac{1 + x}{(1 - x)(1 - 2x)}.$$

Partial fractions give $-2/(1 - x) + 3/(1 - 2x)$, so $e_n = 3 \cdot 2^n - 2$. The initial values are 1 and 4, as required.

12. The coefficients satisfy $t_0 = t_1 = 1$ and $t_n = t_{n-1} + t_{n-2}$. Thus the first six are 1, 1, 2, 3, 5, 8, so the coefficient is eight. It counts ordered tilings of a length-five strip by squares and dominoes, including all possible positions of each piece.

Further reading and scope

These explanations and exercises are original instructional material covering the supplied Unit 3 syllabus. For additional problems and broader context, consult Eric Lehman, F. Thomson Leighton, and Albert R. Meyer, *Mathematics for Computer Science*, through MIT OpenCourseWare. Relevant themes include counting, pigeonhole arguments, recurrences, and ordinary generating functions.

The main recurrence methods here assume linear relations with constant coefficients. Variable-coefficient, nonlinear, and divide-and-conquer recurrences require additional techniques. Generating functions are interpreted formally unless a numerical convergence claim is explicitly made. Keeping these boundaries clear helps identify when a familiar method applies and when a new argument is needed.